

EV CHARGING STATION RESERVATION SYSTEM

Architecture Selection & Justification

Architecture Selection: We chose Layered Architecture for the EV Charging Station Reservation System.

Architectural Choice

The EV Charging Station Reservation System uses Layered Architecture, separating the presentation, business, and data responsibilities. The EV Driver Interface handles user interactions, while business components manage reservations, payments, charging station operations, and the database stores system information.

Reason 1 – Separation of Responsibilities

Layered Architecture separates user interaction from core business operations. Reservation management, connector compatibility, socket availability, and pricing can be handled in the business layer without mixing them with the user interface.

Reason 2 – Easier Maintenance

The system contains different responsibilities such as reservation processing, payment processing, charging station management, and data storage. Separating these responsibilities makes the system easier to understand, maintain, and modify.

Security Advantage

The business layer can validate reservation and payment requests before allowing access to protected data. Access to charging station management functions can also be restricted to authorized station operators.

Performance Benefit

Separating presentation, business processing, and data access allows reservation and charging-station availability requests to be processed in an organized manner. This avoids unnecessary mixing of user-interface operations with database operations.